

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	Medium

Question

Recent measurements of the mass of the *W* boson (a subatomic particle) were notable not only for the mere fact that the particle’s mass differed from expectations but for the _____ of that difference: the measured mass of the *W* boson was seven standard deviations higher than predicted by the standard model of particle physics.

Which choice completes the text with the most logical and precise word or phrase?

Answer

- A. cause
- B. existence
- C. implication
- D. scale

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of the *W* boson. In this context, "scale" means size or extent. The text indicates that recent measurements of the *W* boson were remarkable because they revealed the subatomic particle’s mass to be much greater ("seven standard deviations higher") than had been expected based on a standard model. This context conveys not only that the measurements differed from predictions but also that the extent of the difference was very large.

Choice A is incorrect. Although the text indicates that there was a very large difference between the predicted mass and the actual measured mass of the subatomic particle, it doesn’t explain the cause of, or reason behind, that difference. Choice B is incorrect because it wouldn’t make sense to say that the measurements of the subatomic particle were notable because they differed from expectations and also because a difference existed between the measurements and what had been predicted; the text would be repeating the idea that there was a difference instead of adding to that idea. Choice C is incorrect because the text doesn’t convey any implications of, or consequences of or conclusions drawn from, the fact that the mass of the subatomic particle differed from expectations; the context indicates only that the actual measured mass was much higher than the predicted mass.